

CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

DPP- 01 Discussion Notes



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Sir

#Q. How many simple graphs are possible with 5 vertices and 3 edges?

$${}^5C_3 = \frac{10}{3 \cdot 2} = 10$$

$$= \frac{5 \cdot 4 \cdot 3}{2 \cdot 1} = 10$$

$${}^5C_3 = 10$$

$$= \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$



#Q. A non-directed graph contains 16 edges and all vertices are of degree 2. Then number of vertices in G is _____.

$$\sum d(V) = 2 \cdot E$$

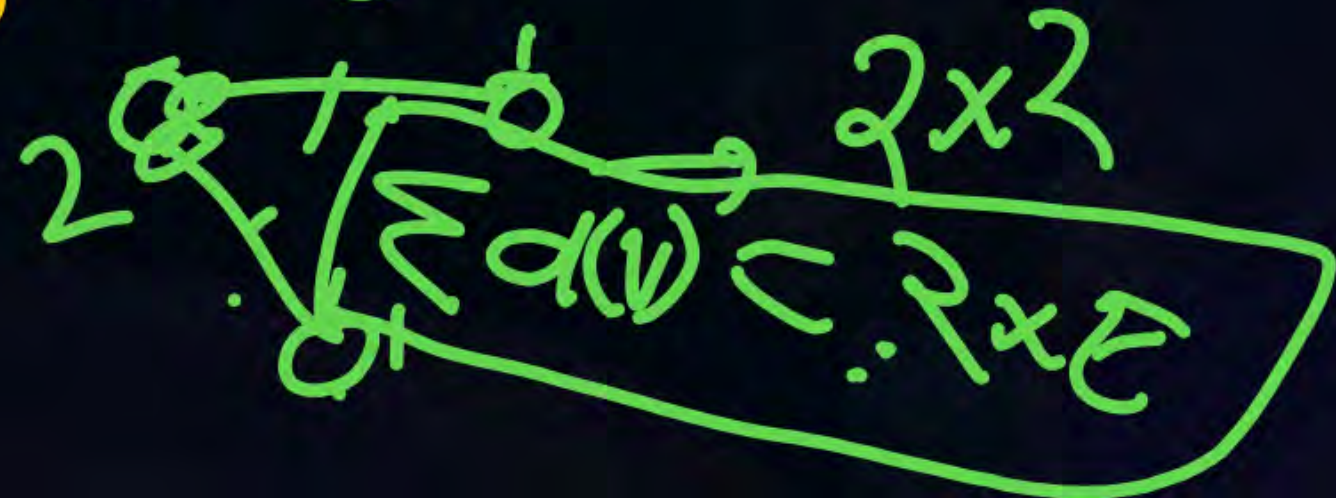
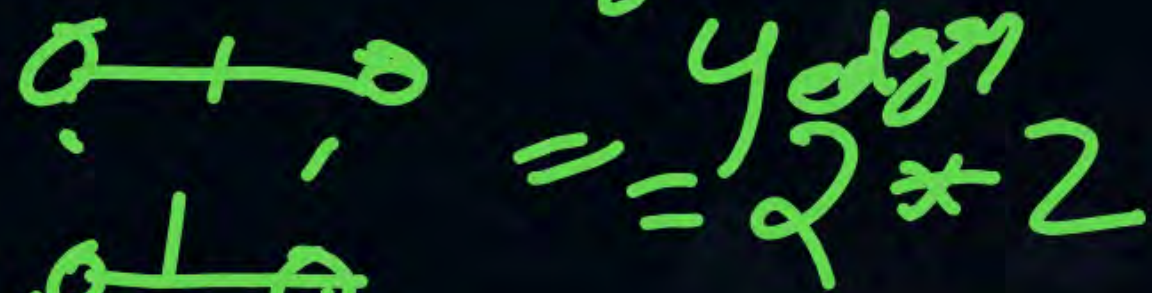
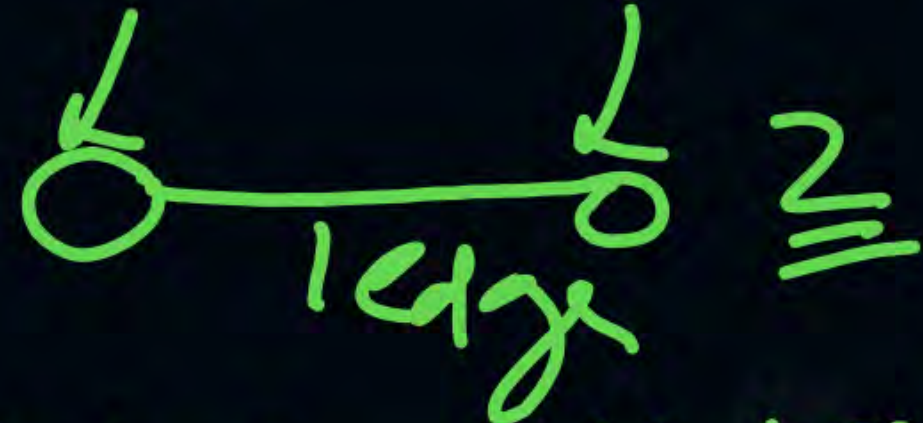
$$= 2 \cdot 16$$

$$V_1 + V_2 + V_3 + \dots + V_n$$

$$2 + 2 + 2 + \dots + 2_n = 2 \cdot 16$$

$$2 \cdot n = 2 \cdot 16$$

$$n = 16$$



#Q. G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at-most 4. Then minimum possible value of n is _____.

Max. $G(V, E)$ $E = 25$ $\sum d(v) = 2 \cdot E$

$\delta \leq \frac{2E}{V} \leq \Delta$

$\frac{2 \cdot 25}{V} \leq 4$

$\frac{25}{2} \leq V \Rightarrow V \geq 12.5$ $V = 13$

#Q. G is an undirected graph with n vertices and 35 edges such that each vertex of G has degree at least 3. Then maximum possible value of n is ____.

$$\delta \leq \frac{2E}{V} \leq \Delta$$

$$\delta \leq \frac{2E}{V}$$

$$3 \leq \frac{2 \cdot 35}{V}$$

$$V \leq \frac{2 \cdot 35}{3} = \frac{70}{3}$$

$$V \leq 23.33 \dots$$

$$V = 23$$

[MSQ]

$G(V, E)$



#Q. If G be a graph with 5 vertices and 7 edges then

$\sum \sum \sum \sum \sum$
10
14

A

$\delta > (G) 2$

B

$\Delta > (G) 4$

C

$\delta \leq (G) 2$

D

$\Delta \leq (G) 2$

2 2 2 2 2 = 10
2 4 2 2 1 3
9

100

$\times$
 $\times$
 $2 \cdot |E|$
 $2 \cdot 7 = 14$

#Q. Find the number of edges of an undirected graph having degree sequence 2, 4, 4, 3, 4, 1.

4, 4, 3, 4, 1?

$G(V, E)$

$$\sum d(v) = 2 \cdot E$$

$$2 + 4 + 4 + 3 + 4 + 1 = 2 \cdot E$$

$$18 = 2 \cdot E$$

$$9 = E$$

#Q. An ordered n-tuple $(d_1, d_2, d_3, \dots, d_n)$ with $d_1 \geq d_2 \geq d_3 \dots \geq d_n$ is called graphic if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, d_3, \dots, d_n$ respectively.

Which of the following is a graphic sequence?

H H Theorem

A 5, 3, 3, 2, 2, 1

C 6, 5, 4, 3, 2, 1

B 2, 1, 1, 1, 1, 1

D 5, 5, 2, 2, 1, 1

2 2 1 1 0

1 0 1 0

0 0 0 0

0 0 1 1 1

1 1 1 0 0

1 0 0 0

[NAT]



#Q. There are 24 routers in an organization. Find the number of cables required to connect them such that each router is connected with exactly 6 other routers.



[NAT]

$$K_n = 2016$$

$$n = 64 - 2 = \underline{62}$$



#Q. Consider a complete graph with size 2016. Suppose after deletion of 2 vertices from the above graph, the modified graph have x number of edges and y number of vertices. Find the value of $x - y$?

$$\frac{n(n-1)}{2} = 2016$$

$$n^2 - n = 2 \cdot 2016$$

$$n^2 - n - 4032 = 0$$

$$\begin{aligned} n &= \frac{n \cdot n - 1}{2} \\ &= \frac{n \cdot n - 1}{2} \end{aligned}$$

$$n^2 - n - 4032 = 0$$

$$n = \frac{-(-1) \pm \sqrt{(-1)^2 + 4 \cdot 1 \cdot (+4032)}}{2 \cdot 1}$$

$$16128$$

$$n = 64$$

$$n = 1 + \sqrt{1 + 16128}$$

$$n = \frac{1 + \sqrt{16129}}{2}$$

$$n = \frac{1 + 127}{2} = \frac{128}{2}$$

$$(64)$$

$$n = \underline{\underline{62}} \leftarrow \textcircled{y=62}$$

$$n = \text{no. of edges}$$

$$= \textcircled{62C_2} \Rightarrow$$

$$nC_2 = \frac{n \cdot n-1}{2}$$

$$\sqrt{2} \sqrt{n-2} = \textcircled{\frac{n(n-1)}{2}}$$

$$= \frac{62 \cdot 61}{2} = 1891 - 62 = \textcircled{1829}$$

$$\begin{array}{r} 61 \\ \times 31 \\ \hline 61 \\ 183 \times \\ \hline 1891 = X \end{array}$$

[MCQ]



#Q. The degree sequences of a simple graph is the sequence of the degree of the nodes in the graph in decreasing order. Which of the following sequences cannot be the degree sequence of any graph?

I. 7,6,5,4,4,3,2,1

II. 6,6,6,6,3,3,2,2

III. 7,6,6,4,4,3,2,2

IV. 8,7,7,6,4,2,1,1

A

I and II

C

IV only

B

III and IV

D

II and IV

Handwritten notes for sequence I: 7, 6, 5, 4, 4, 3, 2, 1. The 5 is circled. Below it, a sequence 4, 3, 3, 2, 1, 0 is written, with 4, 3, 3, 2, 1 underlined. Below that, 3, 2, 2, 1, 0 is written, with 3, 2, 2, 1 underlined. At the bottom, 1, 0, 0, 0 is written, with 1, 0, 0 underlined.

Handwritten notes for sequence II: 6, 6, 6, 6, 3, 3, 2, 2. The sequence is written twice, with the first instance having 6, 6, 6, 6 underlined. Below it, 4, 1, 1, 1, 1 is written, with 4, 1, 1, 1 underlined. At the bottom, 3, 0, 0, 0, 1 is written, with 3, 0, 0 underlined.

DS

THANK - YOU